

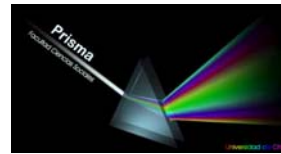
Microscopy, Modeling and Visualization

course spring 2006

Physics of Colored Light

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How physical is Color?



- Physics
- Human Visual System
- Perception
- Computers and color
- Color spaces

Special thanks to Olympus (microscope firm)
from whose site most of the images were taken

The Physics of Light



- Light: em-wave
- Spectrum: color and wavelength
- Light: particle aspect
- – photon
- Light-object interactions:
 - - reflection
 - - refraction
 - - diffraction
 - - absorption/emission

Physics of Light

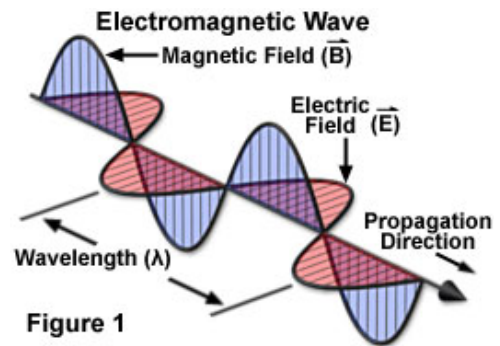
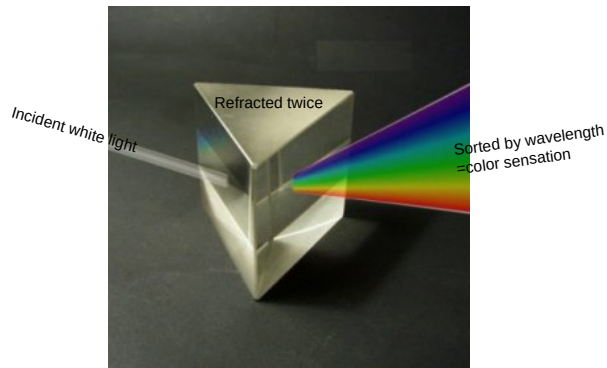


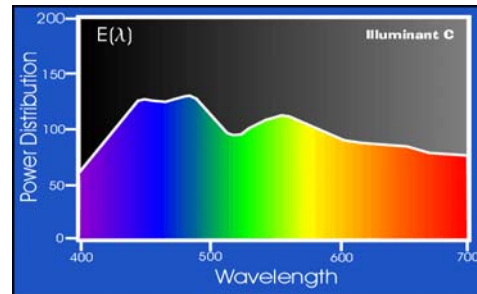
Figure 1

Wavelength and color



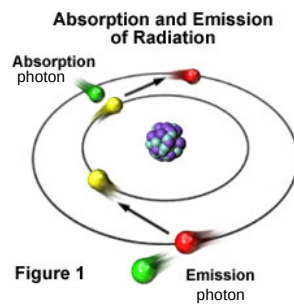
Spectrum

- Light intensity sorted by/function of wavelength:
- How is the energy E of a light source C distributed over λ



Light: particle nature

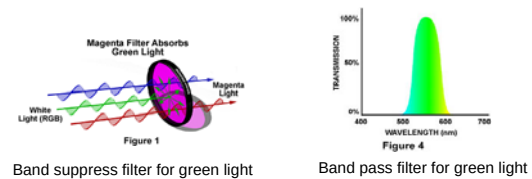
- Interaction with light: Energy package
- Photon: $E = h \cdot \nu$ with $\nu = c/\lambda$
- ν : frequency; h : Planck's constant
- λ : wavelength nanometers
- c : velocity of light



For instance in microscope: incident blue light photon $\rightarrow$ emitted phosphorescence photon

Wavelength filters: absorption, transmission

- In microscopy filters are often used to:
- Let through or block certain wavelength intervals



Object interaction with light: reflection

Reflections From the Surface of Water



Smooth Water Surface

Wavy Water Surface

Figure 1

Reflection angles

Mirror like (specular reflection): equal angles with normal to surface

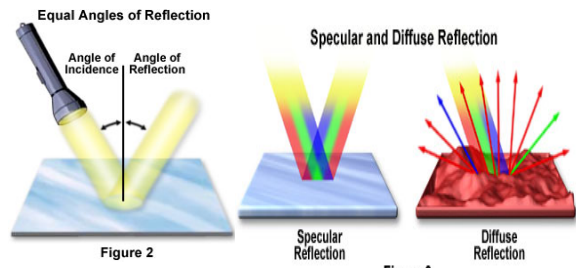


Figure 2

Specular
Reflection

Figure 3

Diffuse
Reflection

Random (diffuse) reflection: highly irregular surface microstructure

Object interaction with light: refraction

Light Refraction by a Glass of Water



Figure 2

Snell's Law and Refractive Index Effects

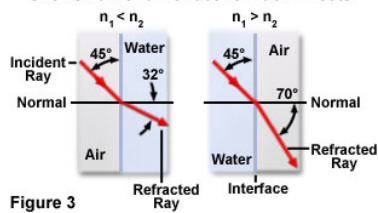


Figure 3

Light moves faster in less dense medium:
Speed difference between media determines refraction

refraction

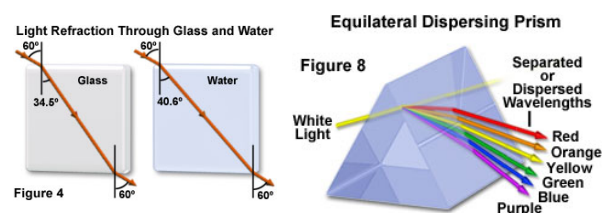


Figure 4

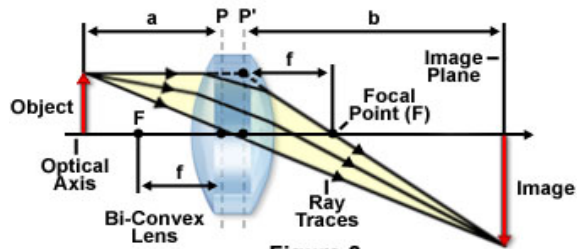
Figure 8

Separated
or
Dispersed
Wavelengths

Red
Orange
Yellow
Green
Blue
Purple

A simple lens

Simple Thin Lens Geometrical Optics



Lens formula: $1/a + 1/b = 1/f$

a =object distance

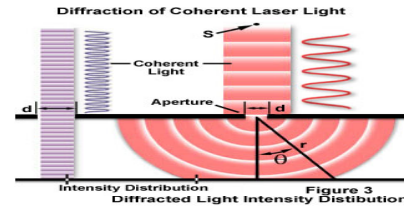
b =image distance

f =focal distance

Figure 2

Image construction: 3 easy rays
 - Undisturbed through center
 - parallel to axis in: through f out
 - through f in: parallel out

Object interaction with light: wave diffraction when slit size $\sim \lambda$



Diameter $\gg \lambda$

Particle model

No bending around edges

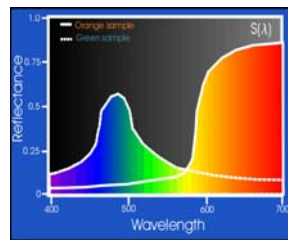
Diameter $\sim \lambda$: wave model



Figure 4

Change of spectrum due to surface reflectance

- Due to object surface reflectance the spectral distribution incident from the lightsource (sun, bulb) will change



Reflectance function $S(\lambda)$

Further change of spectrum due to sensor spectral sensitivity

Absorption Spectra of Human Visual Pigments

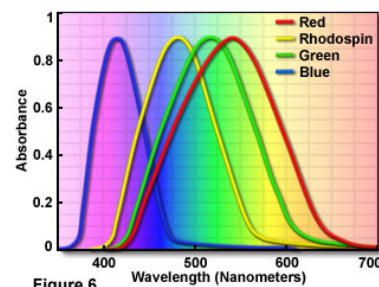


Figure 6

Absorbance as a function of wavelength $A(\lambda)$ for rhodopsin and human cones

Detected light intensity: a game for 4 players



Light source: $E(\lambda)$
 Reflecting surface: $S(\lambda)$
 Absorbing medium: $A(\lambda)$
 Light from object:
 $A(\lambda) \cdot S(\lambda) \cdot E(\lambda)$
 Detector absorbance: $D(\lambda)$
 Intensity from object in
 direction detected:
 $I = \int D(\lambda) \cdot A(\lambda) \cdot S(\lambda) \cdot E(\lambda) d(\lambda)$

The resulting intensity is direction and time dependent

Yet another light characteristic: polarization

Polarization comes in 2 flavours:
 flavour 1 linear polarization in direction perpendicular to propagation;
 2 independent components (horizontal/vertical)

Light Passing Through Crossed Polarizers

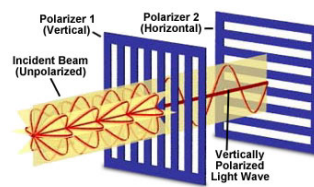


Figure 1

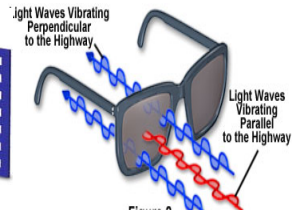
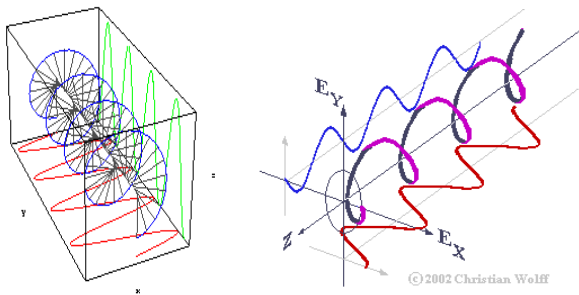


Figure 2

Flavour 2: circular (rotating) polarization; clockwise and/or anti-clockwise

Circular polarization



LCD displays use polarization

Seven-Segment Liquid Crystal Display (LCD)

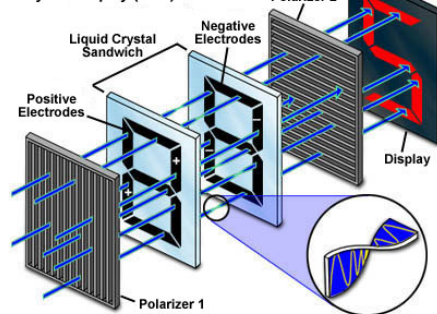
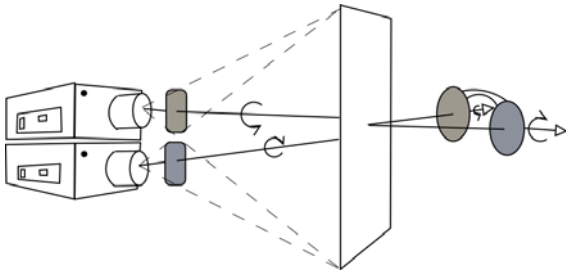
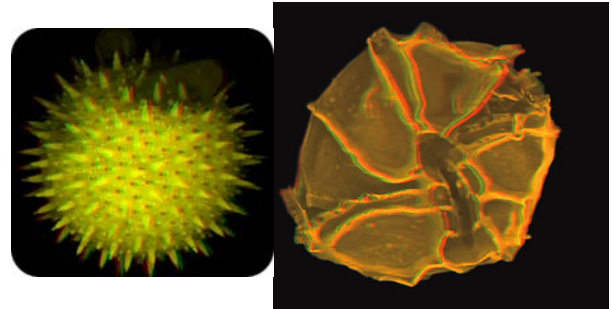


Figure 3

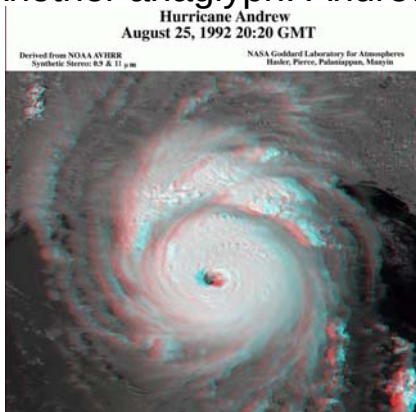
Stereo display with polarized light



Anaglyph: Red-green stereo



Another anaglyph: Andrew



Overview physics of colored light

- Dominant wavelength and color sensation
- Physical interactions
- Inference of surface characteristics due to changes in spectral distribution
- Measured intensity is integrated part of changed light source spectrum
- Physics of light very important for microscopes and cls systems